INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2022/061593

A. CLASSIFICATION OF SUBJECT MATTER
INV. G01N33/68 G01N33/74
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
G01N

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)
EPO-Internal, BIOSIS, CHEM ABS Data, EMBASE, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	MINA HUR ET AL: "Diagnostic and prognostic utilities of multimarkers approach using procalcitonin, B-type natriuretic peptide, and neutrophil gelatinase-associated lipocalin in critically ill patients with suspected sepsis", BMC INFECTIONS DISEASES, BIOMED CENTRAL, LONDON, GB, vol. 14, no. 1, 24 April 2014 (2014-04-24) , page 224, XP021184432, ISSN: 1471-2334, DOI: 10.1186/1471-2334-14-224 abstract; p.7, co. 1; p. 7, par.: "Conclusions"	1-22

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[x] Further documents are listed in the continuation of Box C.

[x] See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
25 August 2022	03/11/2022

Name and mailing address of the ISA/	Authorized officer
European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016	Schalich, Juliane

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Form PCT/ISA/210 (second sheet) (April 2005)

INTERNATIONAL SEARCH REPORT

International application No
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C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	CN 111 208 305 A (YANG YIXUAN) 29 May 2020 (2020-05-29) abstract, claims 1-2	1-22
A	OSTERMANN MARLIES ET AL: "Cardiac Troponin Release is Associated with Biomarkers of Inflammation and Ventricular Dilatation During Critical Illness", SHOCK, vol. 47, no. 6, 1 June 2017 (2017-06-01), pages 702-708, XP055854979, US ISSN: 1073-2322, DOI: 10.1097/SHK.0000000000000811 the whole document	1-22
X	HEILMANN EVA ET AL: "Biomarkers of Infection: Are They Useful in the ICU?", SEMINARS IN RESPIRATORY AND CRITICAL CARE MEDICINE, vol. 40, no. 04, 1 August 2019 (2019-08-01), pages 465-475, XP055955145, US ISSN: 1069-3424, DOI: 10.1055/s-0039-1696689 Fig. 2; p. 468, co. 1; Table 1	1,2,4-9, 17,18
X	SIEGLER B H ET AL: "Use of biomarkers in sepsis", ANAESTHESIST, SPRINGER VERLAG, DE, vol. 63, no. 8, 9 July 2014 (2014-07-09), pages 678-690, XP035401998, ISSN: 0003-2417, DOI: 10.1007/S00101-014-2347-2 [retrieved on 2014-07-09]	1-9,17, 18,21,22
Y	Table 1; Fig. 3; p. 684, par.: Kardiale Troponine	10-16
X	CN 109 841 278 A (CHANG GUNG MEDICAL FOUND LINKOU CHANG GUNG MEMORIAL HOSPITAL ET AL.) 4 June 2019 (2019-06-04)	19
Y	claims 1, 6	10-16

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Box No. II Observations where certain claims were found unsearchable (Continuation of item 2 of first sheet)

This international search report has not been established in respect of certain claims under Article 17(2)(a) for the following reasons:

1. ☐ Claims Nos.:
because they relate to subject matter not required to be searched by this Authority, namely:

2. ☐ Claims Nos.:
because they relate to parts of the international application that do not comply with the prescribed requirements to such an extent that no meaningful international search can be carried out, specifically:

3. ☐ Claims Nos.:
because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a).

Box No. III Observations where unity of invention is lacking (Continuation of item 3 of first sheet)

This International Searching Authority found multiple inventions in this international application, as follows:

see additional sheet

1. ☐ As all required additional search fees were timely paid by the applicant, this international search report covers all searchable claims.

2. ☐ As all searchable claims could be searched without effort justifying an additional fees, this Authority did not invite payment of additional fees.

3. ☐ As only some of the required additional search fees were timely paid by the applicant, this international search report covers only those claims for which fees were paid, specifically claims Nos.:

4. ☒ No required additional search fees were timely paid by the applicant. Consequently, this international search report is restricted to the invention first mentioned in the claims;; it is covered by claims Nos.:

1-22 (partially)

Remark on Protest

☐ The additional search fees were accompanied by the applicant's protest and, where applicable, the payment of a protest fee.

☐ The additional search fees were accompanied by the applicant's protest but the applicable protest fee was not paid within the time limit specified in the invitation.

☐ No protest accompanied the payment of additional search fees.

FURTHER INFORMATION CONTINUED FROM PCT/ISA/ 210

This International Searching Authority found multiple (groups of) inventions in this international application, as follows:

1. claims: 1-22(partially)

Determination of a marker panel comprising PCT, a cardiac troponin and bilirubin for assessing a subject with suspected infection.

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2. claims: 1-22(partially)

Determination of a marker panel comprising PCT, a cardiac troponin and Aspartate aminotransferase and/or Alanine aminotransferase for assessing a subject with suspected infection.

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3. claims: 1-22(partially)

Determination of a marker panel comprising PCT, a cardiac troponin and haptoglobin for assessing a subject with suspected infection.

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4. claims: 1-22(partially)

Determination of a marker panel comprising PCT, a cardiac troponin and ESM-1 for assessing a subject with suspected infection.

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5. claims: 1-22(partially)

Determination of a marker panel comprising PCT, a cardiac troponin and Creatinine for assessing a subject with suspected infection.

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6. claims: 1-22(partially)

Determination of a marker panel comprising PCT and creatinine for assessing a subject with suspected infection.

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7. claims: 1, 3-9, 11-13, 15(all partially)

Determination of a marker panel comprising PCT and a BNP-type peptide for assessing a subject with suspected infection.

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8. claims: 1-22(partially)

Determination of a marker panel comprising PCT and sTREM1

FURTHER INFORMATION CONTINUED FROM PCT/ISA/ 210

for assessing a subject with suspected infection.

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9. claims: 1, 3-10, 12, 15, 17, 19, 21, 22(all partially)

Determination of a marker panel comprising PCT and ESM-1 for assessing a subject with suspected infection.

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10. claims: 1, 3-10, 12, 15, 17, 19, 21, 22(all partially)

Determination of a marker panel comprising PCT and haptoglobin for assessing a subject with suspected infection.

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11. claims: 1, 3-10, 12, 15, 17, 19, 21, 22(all partially)

Determination of a marker panel comprising PCT and Heparin Binding Protein (HBP) for assessing a subject with suspected infection.

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12. claims: 1, 3-10, 12, 15, 17, 19, 21, 22(all partially)

Determination of a marker panel comprising PCT and Aspartate aminotransferase for assessing a subject with suspected infection.

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
PCT/EP2022/061593

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
CN 111208305 A	29-05-2020	NONE	
CN 109841278 A	04-06-2019	NONE	